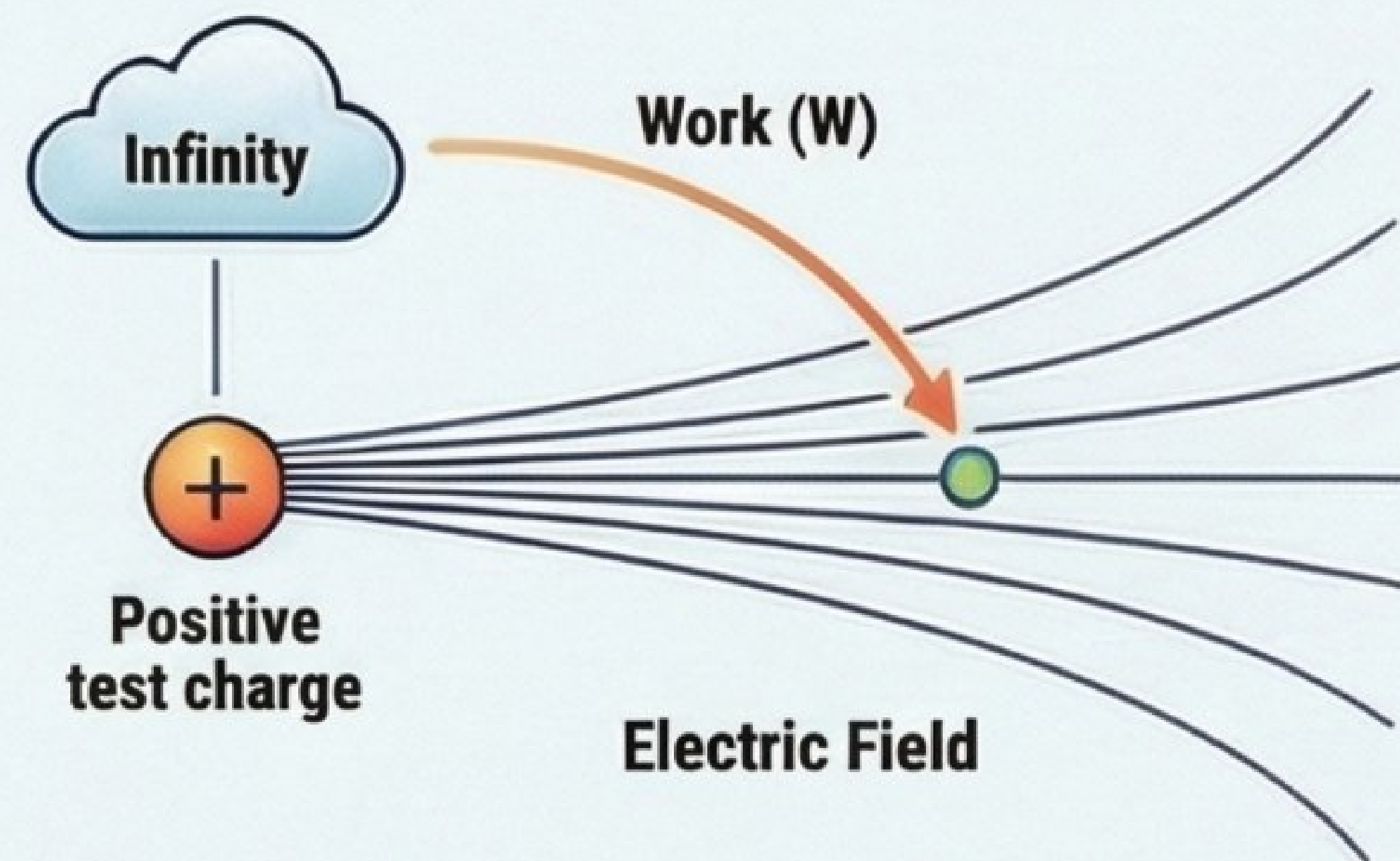


Mastering Electric Potential & Capacitance: A Visual Revision Guide

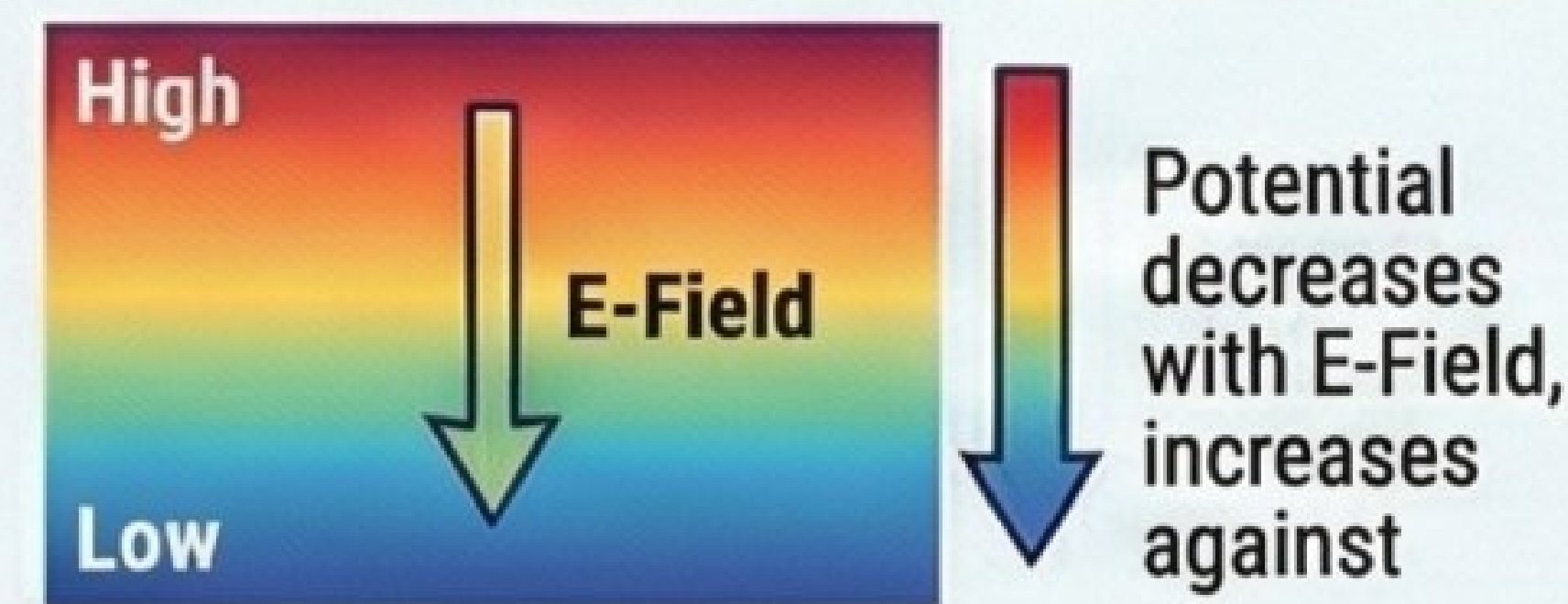
Fundamentals of Electric Potential (V)



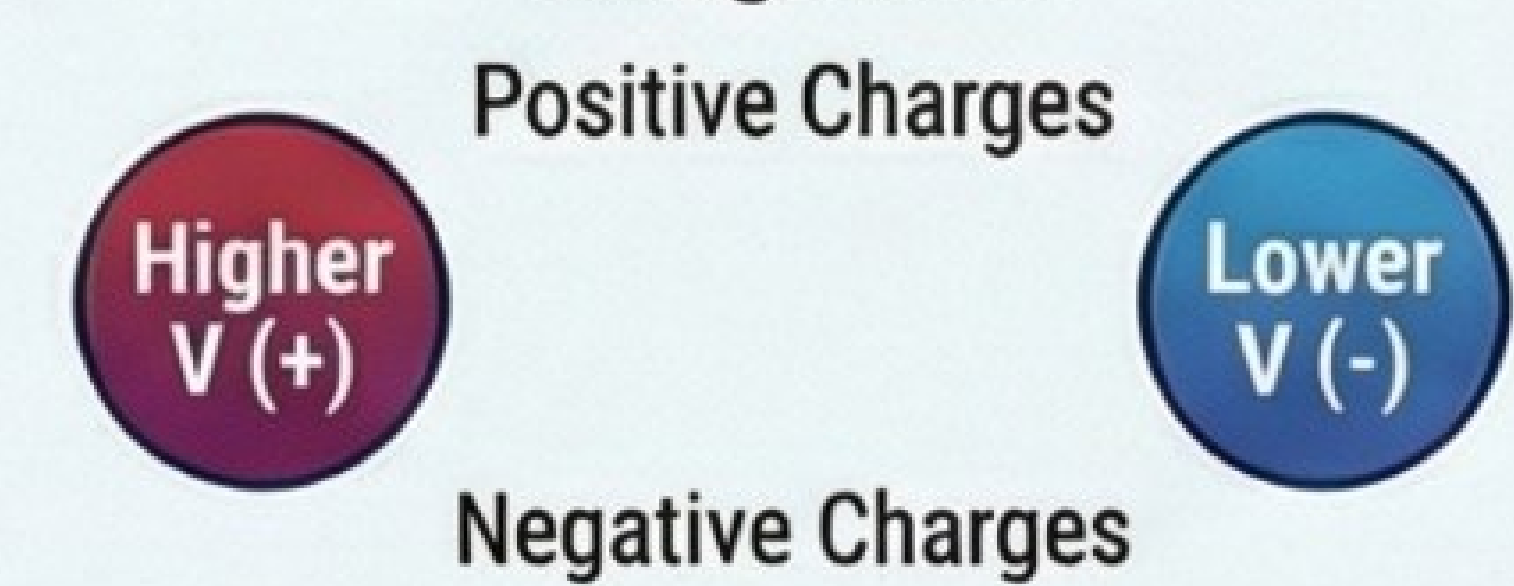
$$V = \frac{W}{q_0}$$

1 Volt
1 Joule of work
per 1 Coulomb

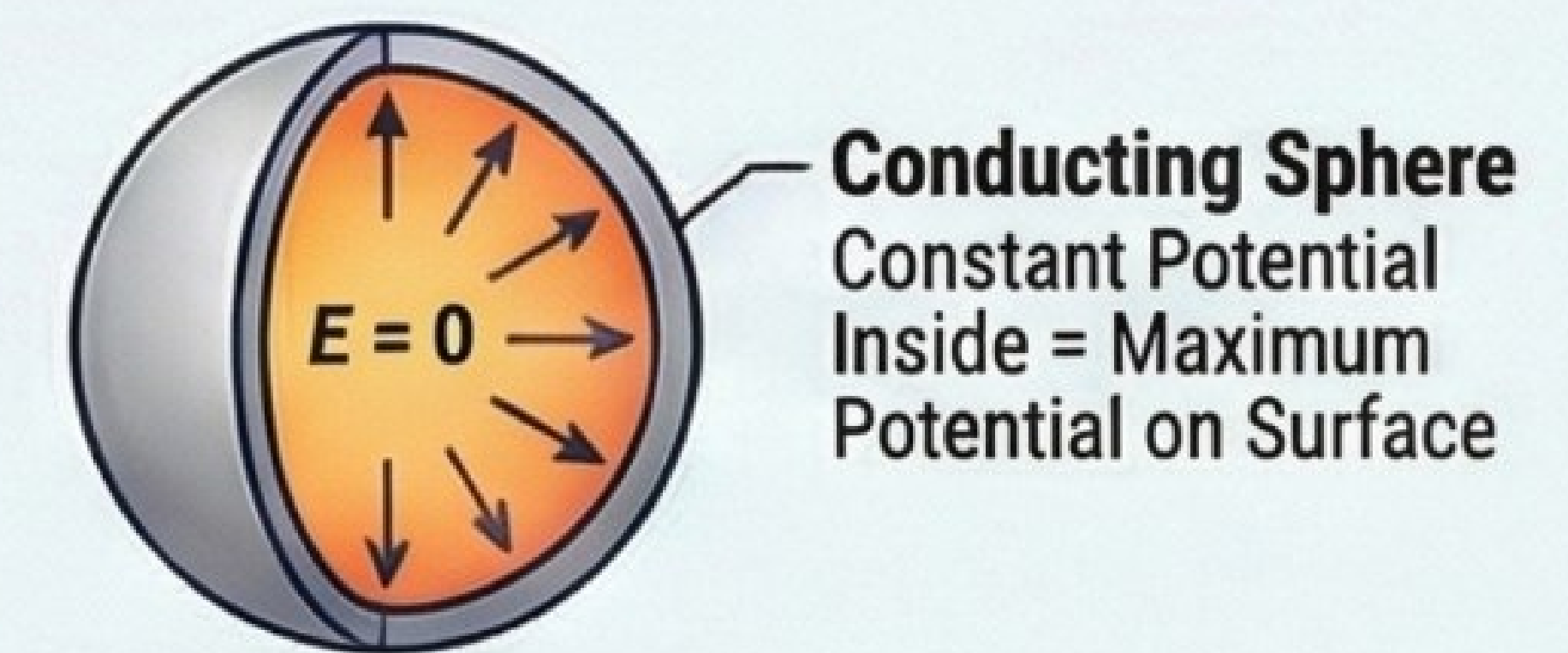
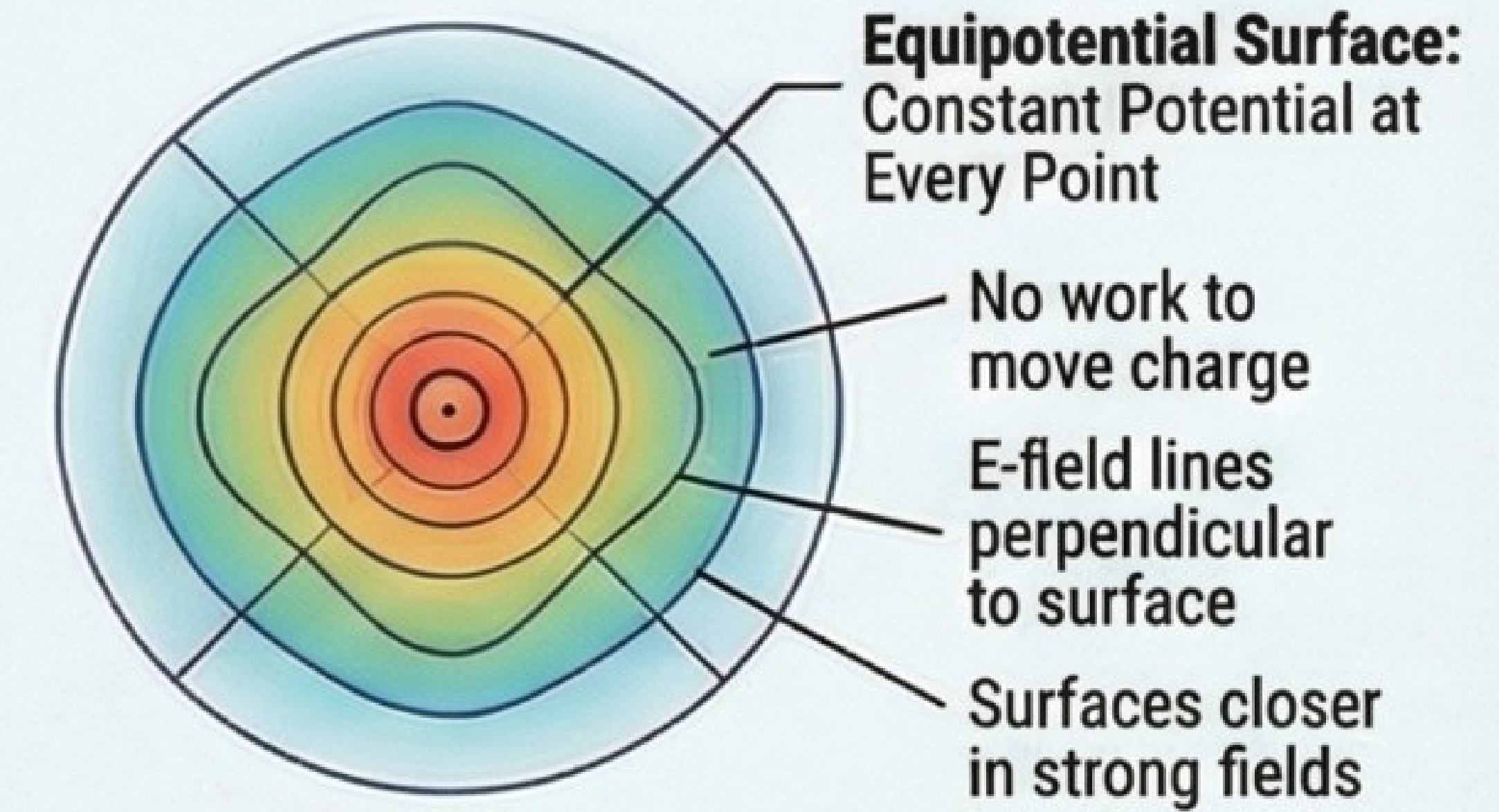
Potential and Electric Field Direction



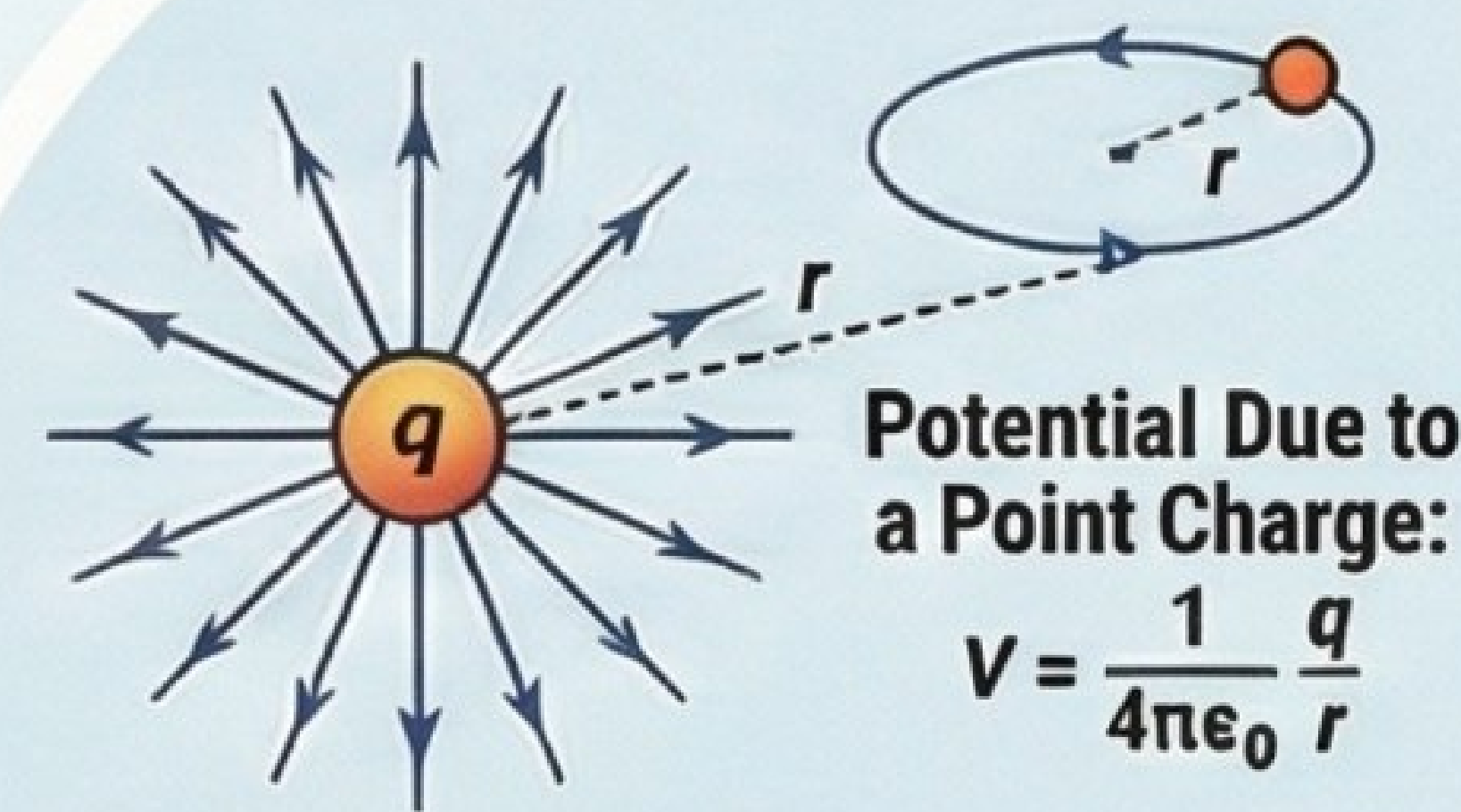
Potential Difference Determines Charge Flow



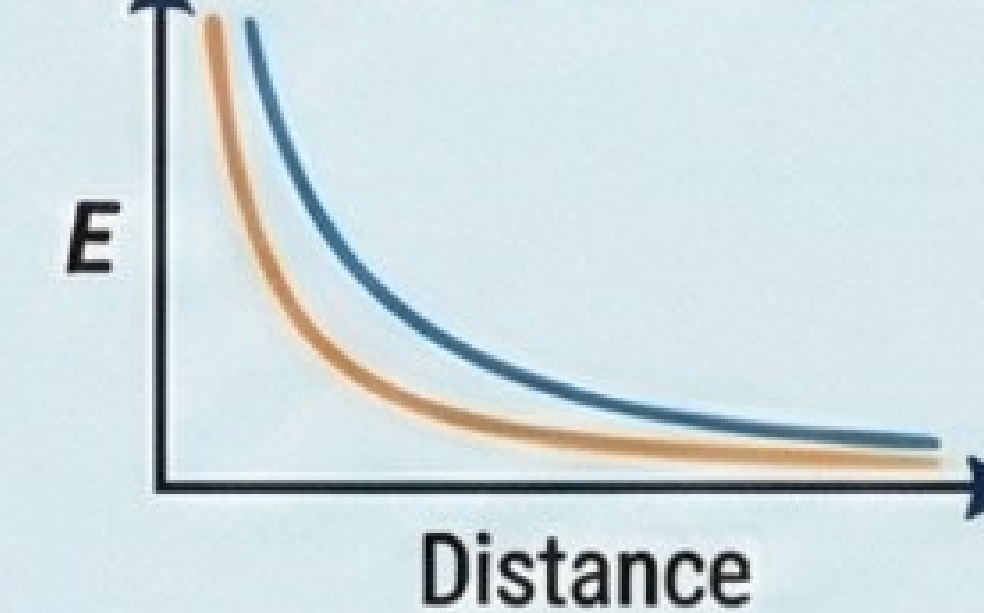
Equipotential Surfaces & Conductors



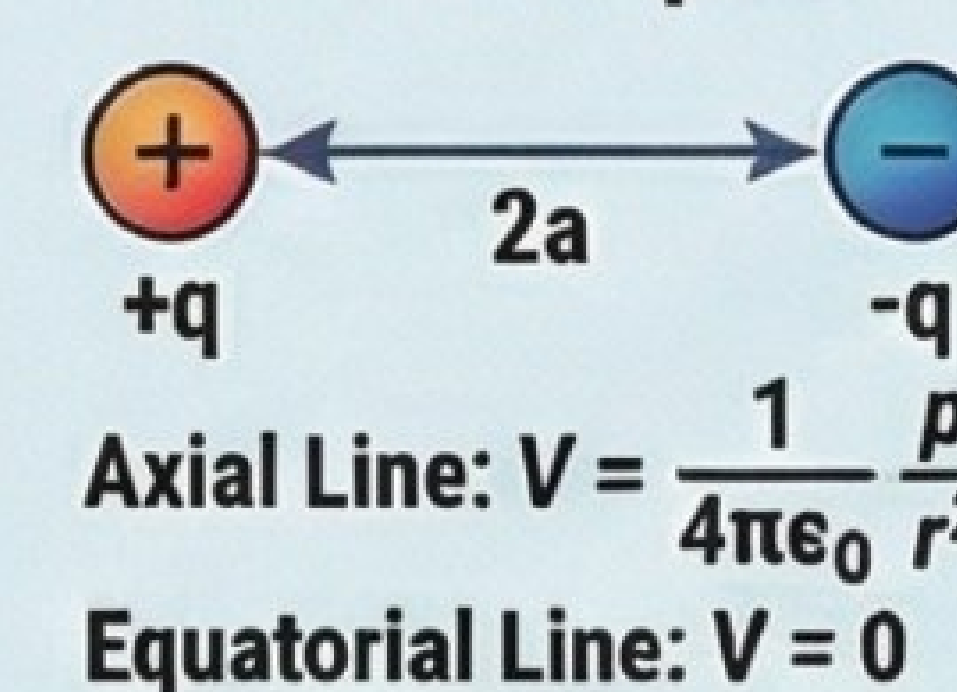
Potential Calculations & Key Formulas



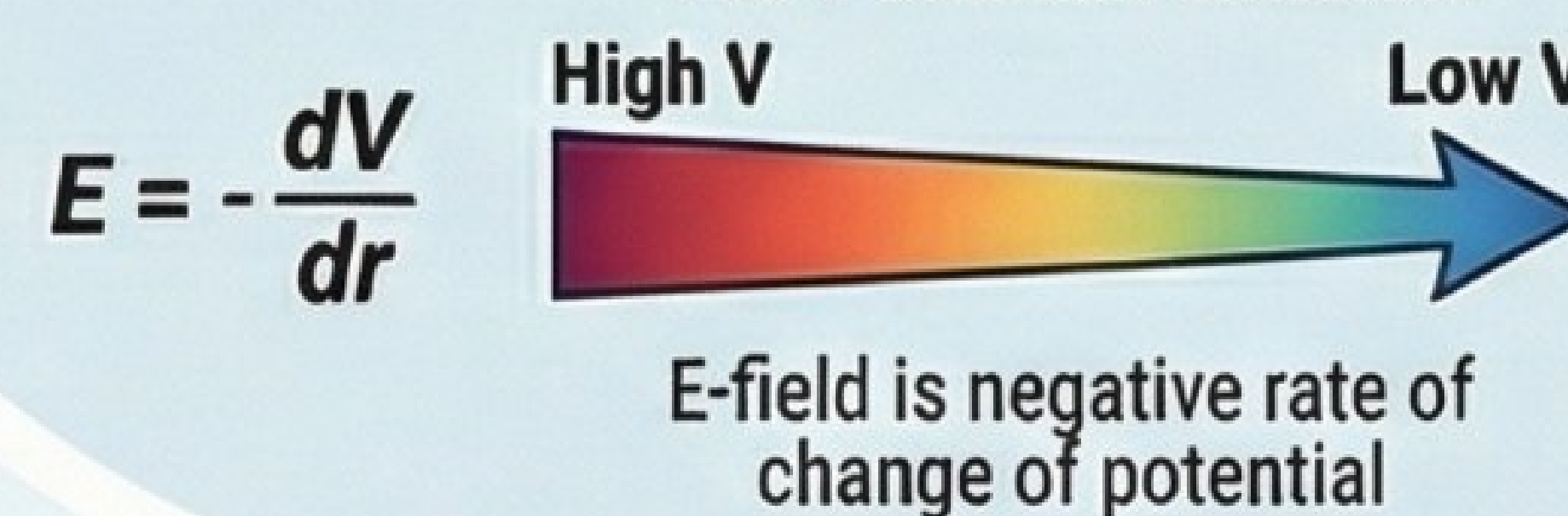
$E (E \propto 1/r^2)$ vs. $V (V \propto 1/r)$
with Distance



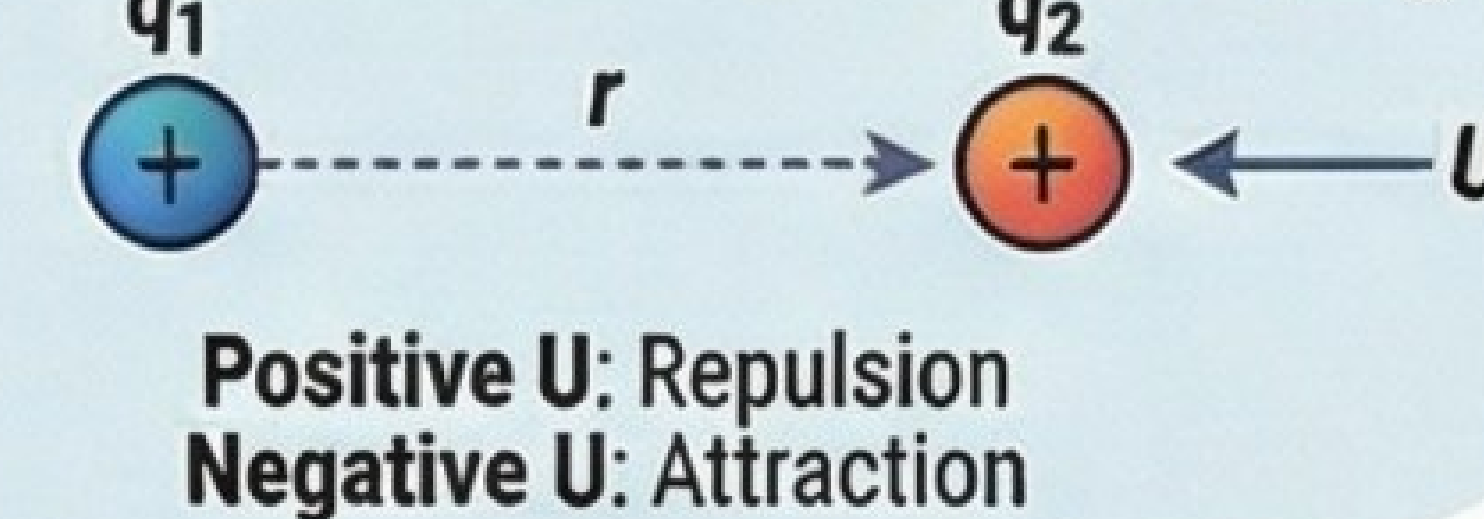
Electric Dipole



The Potential Gradient



Electric Potential Energy (U) = $\frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r}$



Capacitors and Capacitance

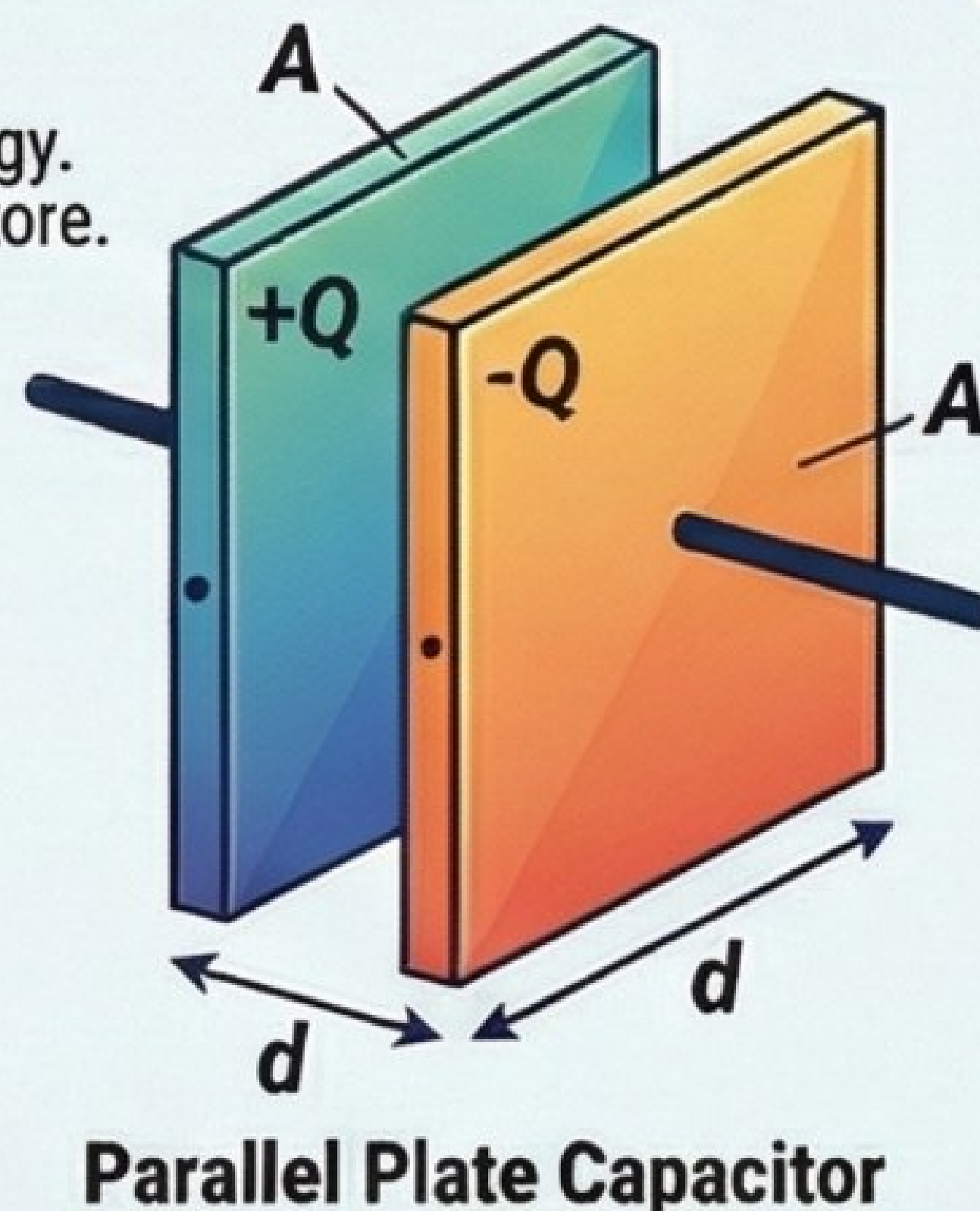
What is a Capacitor?

Device to store electrical energy.
Capacitance (C) is ability to store.

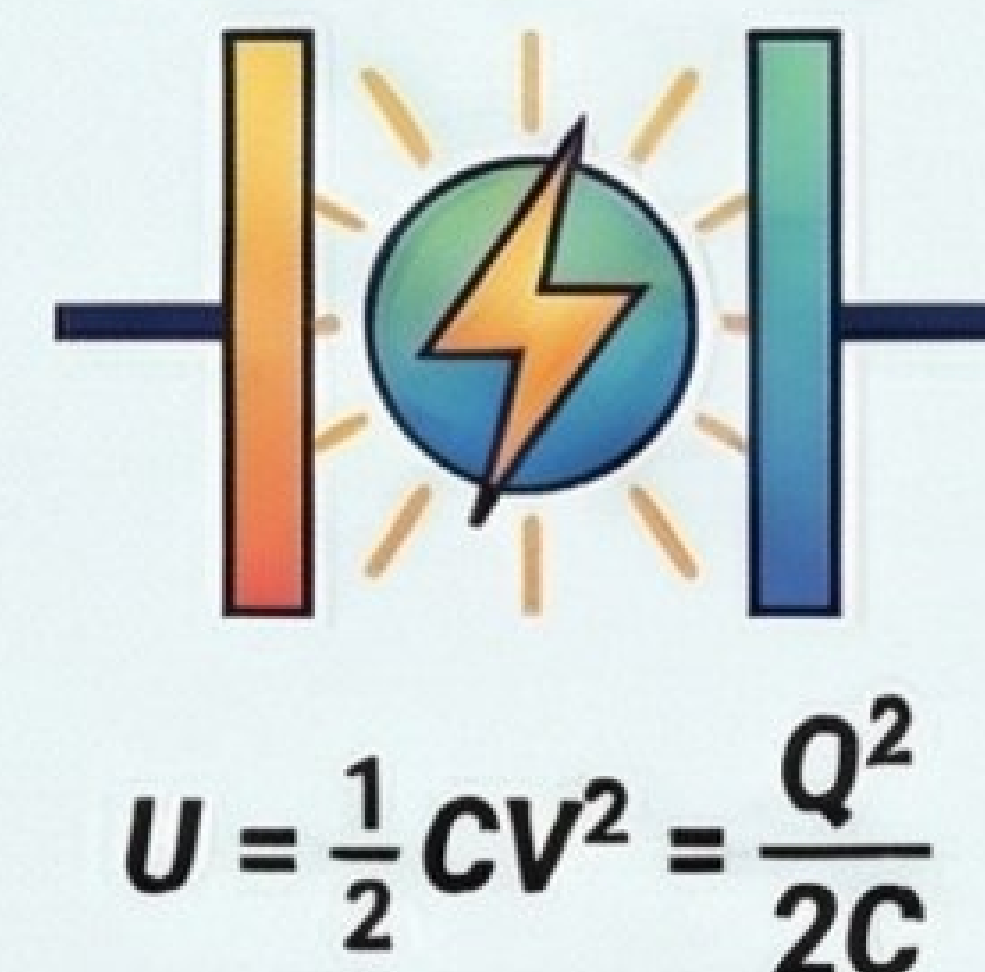
$$C = \frac{Q}{V} \text{ Farad (F)}$$

Depends on Geometry (A, d) and Medium
Not on Q or V

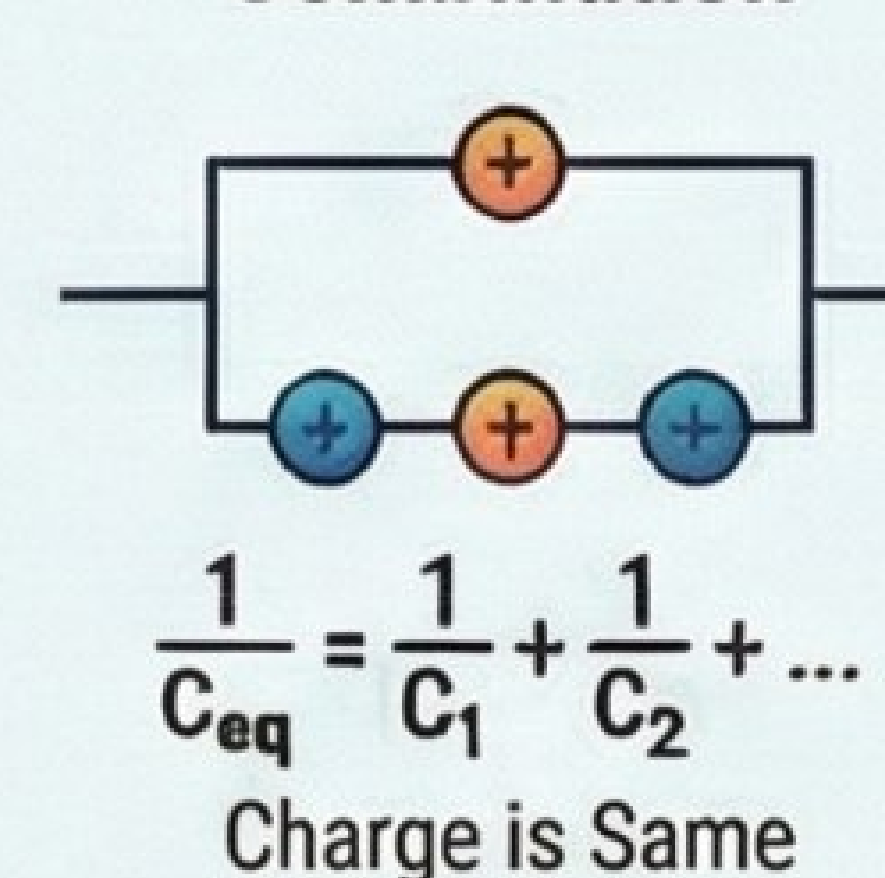
$$C = \frac{\epsilon_0 A}{d}$$



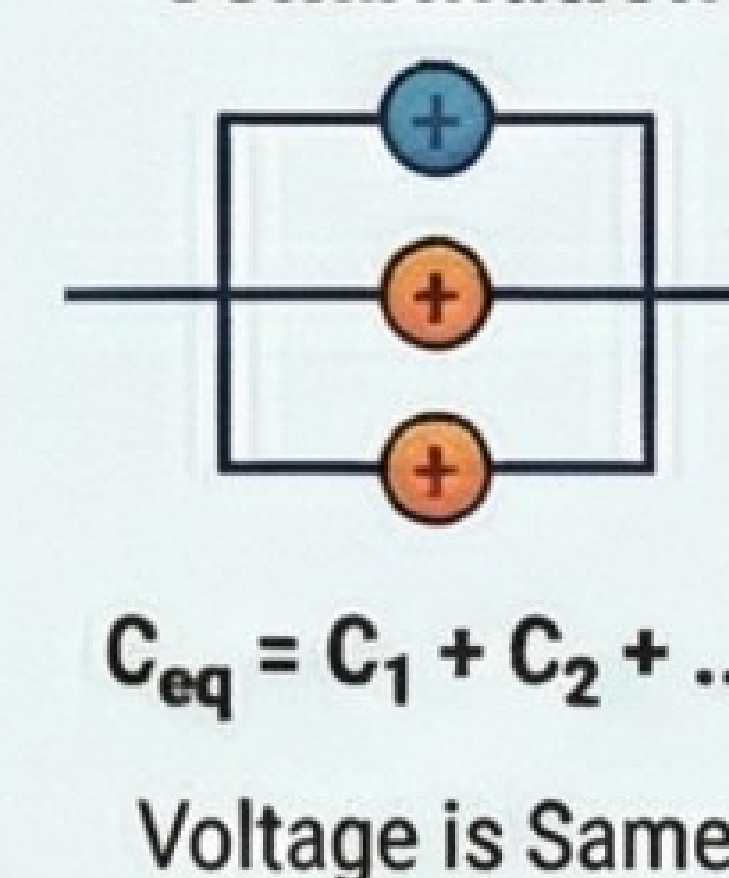
Energy Stored in a Capacitor



Series Combination



Parallel Combination



Combining Capacitors & The Effect of Dielectrics

Dielectric (Constant K)

- Charge (Q)
- Electric Field (E)
- Potential (V)
- Capacitance (C)
- Energy (U)

	Battery is Disconnected	Battery Remains Connected
Charge (Q)	Stays Constant (Q_0)	Increases by K ($K \cdot Q_0$)
Electric Field (E)	Decreases by K ($\frac{E_0}{K}$)	Stays Constant (E_0)
Potential (V)	Decreases by K ($\frac{V_0}{K}$)	Stays Constant (V_0)
Capacitance (C)	Increases by K ($K \cdot C_0$)	Increases by K ($K \cdot C_0$)
Energy (U)	Decreases by K ($\frac{U_0}{K}$)	Increases by K ($K \cdot U_0$)